Tutorial Questions (Set-VIII)

Q1 Let (X, d) be a compact metric space and $C(X) = \{f \colon X \to \mathbb{R} \mid f \text{ is continuous}\}$ be the space of continuous functions on X.

(i) Show that $C(X)$ is an $\mathbb{R}$-algebra under pointwise operations.

(ii) Show that $\|f\|_\infty = \sup\{|f(x)| : x \in X\}$ for $f \in C(X)$ defines a norm-function on $C(X)$.

(iii) Show that $C(X)$ is a complete metric space under the sup-metric given by

$$d_\infty(f, g) = \|f - g\|_\infty \quad \text{for} \quad f, g \in C(X).$$

Q2 Let (X, d) be a compact metric space and $\mathcal{E}$ be an equicontinuous family of functions on X, i.e., $\mathcal{E} \subseteq C(X)$ and $\mathcal{E}$ equicontinuous. Prove the following statements.

(i) If $\mathcal{E}$ is pointwise bounded, then $\mathcal{E}$ is (uniformly) bounded subset of $C(X)$.

(ii) If $(f_n)_{n\in\mathbb{N}}$ is a sequence in $\mathcal{E}$ such that $f_n \to f$ converges pointwise on X, then $f_n \to f$ converges uniformly on X.

(iii) (Arzela - Ascoli Theorem) If $\mathcal{E}$ is closed, pointwise bounded and equicontinuous, then $\mathcal{E}$ is a compact subset of $C(X)$. Further, converse also holds.

(iv) The closed unit ball $\mathbb{B} = \{f \in C[0, 1] : \|f\|_\infty \leq 1\}$ in $C[0, 1]$ is noncompact. In fact, the sequence $\{f_n \in \mathbb{B} : n \in \mathbb{N}\}$ has no uniformly convergent subsequence, where a) $f_n(x) = x^n$, b) $f_n(x) = \frac{x^2}{x^2+(1-nx)^2}$, c) $f_n(x) = \begin{cases} nx, & 0 \leq x \leq \frac{1}{n}, \\ 1, & \frac{1}{n} \leq x \leq 1. \end{cases}$

Q3 (i) (**Optional**) Let $f \in C[0, 1]$. For each $n \in \mathbb{N}$, the sum

$$P_n(x) = \sum_{k=0}^{n} \binom{n}{k} \ f\left(\frac{k}{n}\right) \ x^k \ (1-x)^{n-k} \ ; \quad \text{where } \binom{n}{k} = \frac{n!}{k!(n-k)!},$$

is called the *Bernstein polynomials* associated with f.

(a) For $0 \leq k \leq n$, let $r_k(x) = \binom{n}{k} \ x^k \ (1 - x)^{n-k}$. Prove the identities

$$\sum_{k=0}^{n} r_k(x) = 1 \text{ and } \quad \sum_{k=0}^{n} (k - nx)^2 \ r_k(x) = nx(1 - x).$$

(b) Show that $P_n \to f$ converges uniformly on $[0, 1]$. (This gives another proof of the Weierstrass approximation theorem.)

(ii) Let $f \in C[0, 1]$ and $\int_0^1 f(x) \ x^n \, dx = 0$ for $n = 0, 1, 2, \ldots$. Show that $f \equiv 0$.

(iii) Put $P_0(x) \equiv 0$ and define $P_{n+1}(x) = P_n(x) + \frac{x^2 - P_n(x)}{2}$ for $n = 0, 1, 2, \ldots$. Show that $(P_n)_{n\in\mathbb{N}}$ is a sequence of polynomials such that $P_n(x) \to |x|$ converges uniformly on $[-1, 1]$.

Q4 Let $\mathcal{A}$ be a subalgebra of $C[0,1]$ and $\overline{\mathcal{A}}$ be the closure of $\mathcal{A}$ in $C[0,1]$. ($\overline{\mathcal{A}}$ is called the *uniform closure* of $\mathcal{A}$.)

(i) Show that $\overline{\mathcal{A}}$ is also a subalgebra of $C[0,1]$.

(ii) If $\mathcal{A}$ separates points of $[0,1]$ and $\mathcal{A}$ vanishes at no point of $[0,1]$, then for distinct points $x_1, x_2 \in [0,1]$ and $c_1, c_2 \in \mathbb{R}$, show that there exists $f \in \mathcal{A}$ such that $f(x_1) = c_1$ and $f(x_2) = c_2$.

(iii) If $f \in \mathcal{A}$ (or $\overline{\mathcal{A}}$), then show that $|f| \in \mathcal{A}$ (resp. $|f| \in \overline{\mathcal{A}}$) .

(iv) If $f, g \in \overline{\mathcal{A}}$, then show that $\max(f,g) \in \overline{\mathcal{A}}$ and $\min(f,g) \in \overline{\mathcal{A}}$.

(v) (**Optional**) Let (X, d) be a compact metric space and let $\mathcal{A}$ be a subalgebra of $C(X)$. If $\mathcal{A}$ separates points on X and if $\mathcal{A}$ vanishes at no point of X, then the uniform closure $\overline{\mathcal{A}} = C(X)$. (This is known as the Stone-Weierstrass theorem.)

Q5 (**Optional**) Let (X, d) be a metric space (not necessarily compact) and $C_b(X)$ be the space of real-valued bounded continuous functions on X. We have seen that $C_b(X)$ is a complete metric space under the sup-metric $d_\infty(f,g) = \|f - g\|_\infty$ for $f, g \in C_b(X)$, induced by the sup norm $\|\cdot\|_\infty$. Fix a point $a \in X$ and for each $p \in X$, define $f_p \colon X \to \mathbb{R}$ by $f_p(x) = d(x,p) - d(x,a)$ for $x \in X$.

(i) Show that $f_p \in C_b(X)$ and $\|f_p\|_\infty \le d(a,p)$.

(ii) Let $\Phi \colon X \to C_b(X)$ be the map given by $\Phi(p) = f_p$ for $p \in X$. Then show that Φ is an isometery, i.e., $d(p,q) = \|f_p - f_q\|_\infty$ for any $p, q \in X$.

(iii) Let $Y = \overline{\Phi(X)}$ be the closure of the image $\Phi(X)$ in $C_b(X)$. Then show that Y is a complete metric space.

(Thus every metric space (X, d) can be embedded in a complete metric space Y as an isometrically isomorphic dense subspace. The metric space Y is called the *completion* of the metric space X and it is essentially unique.)